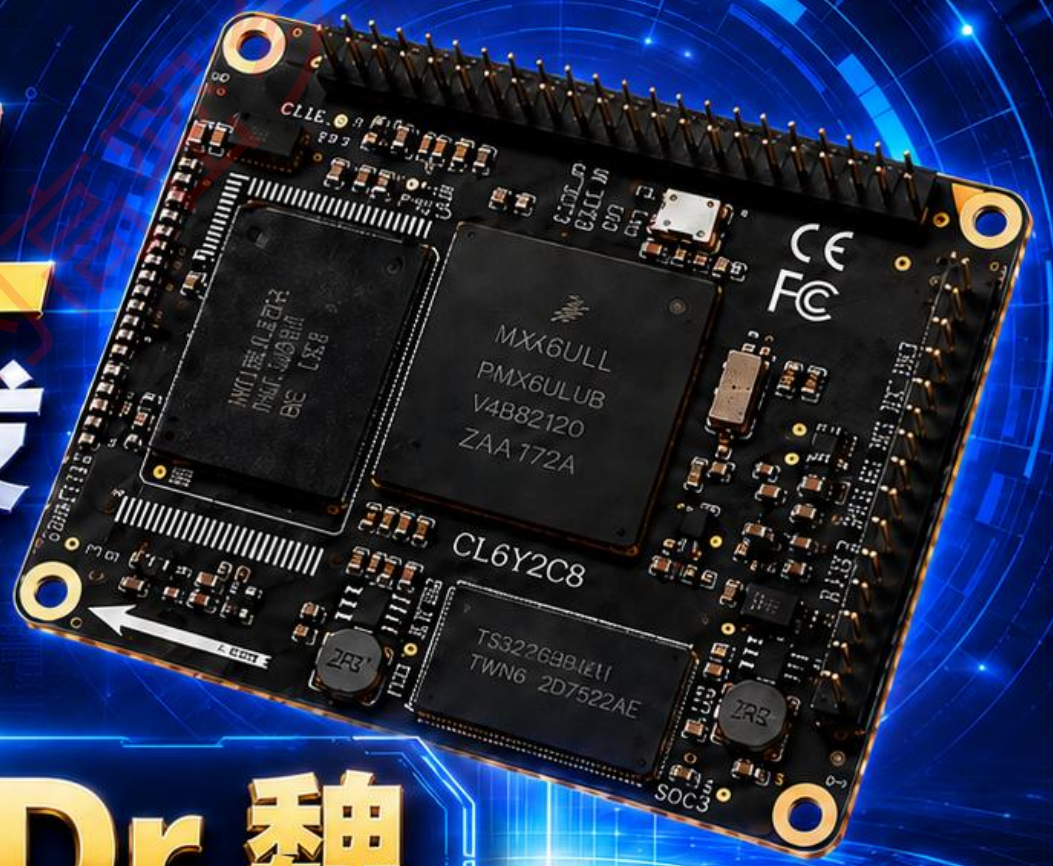
全国最好的

I.MX6ULL

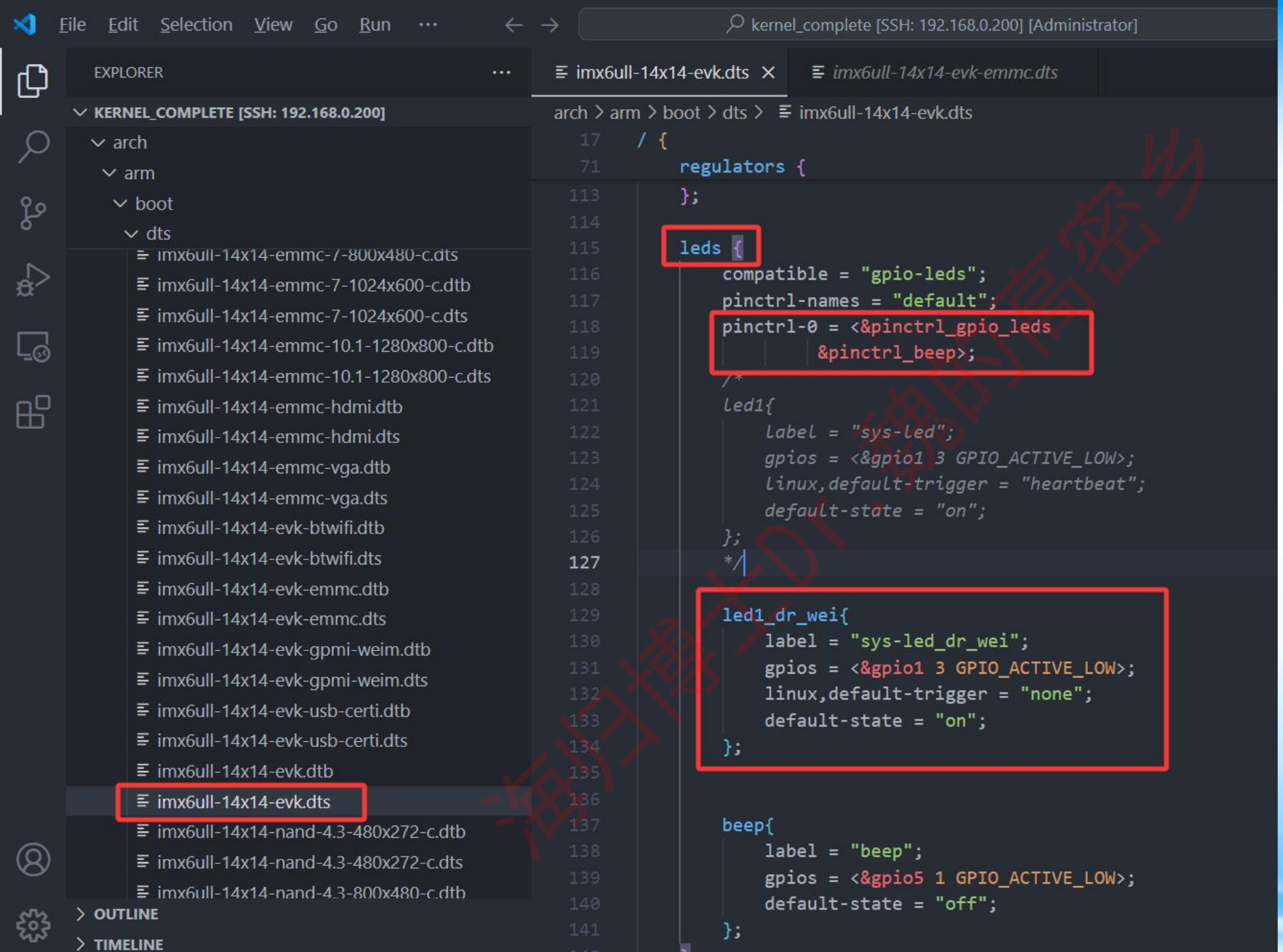
系统移植与驱动开发

铂金课程



主讲人：海归博士Dr.魏

bilibili B站搜索：海归博士Dr.魏的高密乡-技术、教育、哲学与生活的自留地


File Edit Selection View Go Run
kernel_complete [SSH: 192.168.0.200] [Administrator]
EXPLORER
KERNEL_COMPLETE [SSH: 192.168.0.200]
arch
arm
boot
dts
imx6ull-14x14-emmc-7-800x480-c.dts
imx6ull-14x14-emmc-7-1024x600-c.dtb
imx6ull-14x14-emmc-7-1024x600-c.dts
imx6ull-14x14-emmc-10.1-1280x800-c.dtb
imx6ull-14x14-emmc-10.1-1280x800-c.dts
imx6ull-14x14-emmc-hdmi.dtb
imx6ull-14x14-emmc-hdmi.dts
imx6ull-14x14-emmc-vga.dtb
imx6ull-14x14-emmc-vga.dts
imx6ull-14x14-evk-btwifi.dtb
imx6ull-14x14-evk-btwifi.dts
imx6ull-14x14-evk-emmc.dtb
imx6ull-14x14-evk-emmc.dts
imx6ull-14x14-evk-gpmi-weim.dtb
imx6ull-14x14-evk-gpmi-weim.dts
imx6ull-14x14-evk-usb-certi.dtb
imx6ull-14x14-evk-usb-certi.dts
imx6ull-14x14-evk.dtb
imx6ull-14x14-evk.dts
imx6ull-14x14-nand-4.3-480x272-c.dtb
imx6ull-14x14-nand-4.3-480x272-c.dts
imx6ull-14x14-nand-4.3-800x480-c.dtb
OUTLINE
TIMELINE
imx6ull-14x14-evk.dts
imx6ull-14x14-evk-emmc.dts
arch > arm > boot > dts > imx6ull-14x14-evk.dts
17 / {
71 regulators {
113 };
114
115 leds {
116 compatible = "gpio-leds";
117 pinctrl-names = "default";
118 pinctrl-0 = <&pinctrl_gpio_leds
119 &pinctrl_beep>;
120 /*
121 led1{
122 label = "sys-led";
123 gpios = <&gpio1 3 GPIO_ACTIVE_LOW>;
124 linux,default-trigger = "heartbeat";
125 default-state = "on";
126 };
127 */
128
129 led1_dr_wei{
130 label = "sys-led_dr_wei";
131 gpios = <&gpio1 3 GPIO_ACTIVE_LOW>;
132 linux,default-trigger = "none";
133 default-state = "on";
134 };
135
136
137 beep{
138 label = "beep";
139 gpios = <&gpio5 1 GPIO_ACTIVE_LOW>;
140 default-state = "off";
141 };

Welcome

led.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#define LED_BRIGHTNESS "/sys/class/leds/sys-led_dr_wei/brightness"
int main(void)
{
    int fd_brightness;

    /* 打开 brightness */
    fd_brightness = open(LED_BRIGHTNESS, O_RDWR);
    if(fd_brightness < 0)
    {
        perror("open brightness error");
        exit(-1);
    }
    while(1)
    {
        write(fd_brightness, "1", 1);   // LED亮
        sleep(1);

        write(fd_brightness, "0", 1);   // LED灭
        sleep(1);
    }
    close(fd_brightness);
    return 0;
}
```

海贝博士Dr·魏的高级

① menuconfig：决定“编不编驱动”

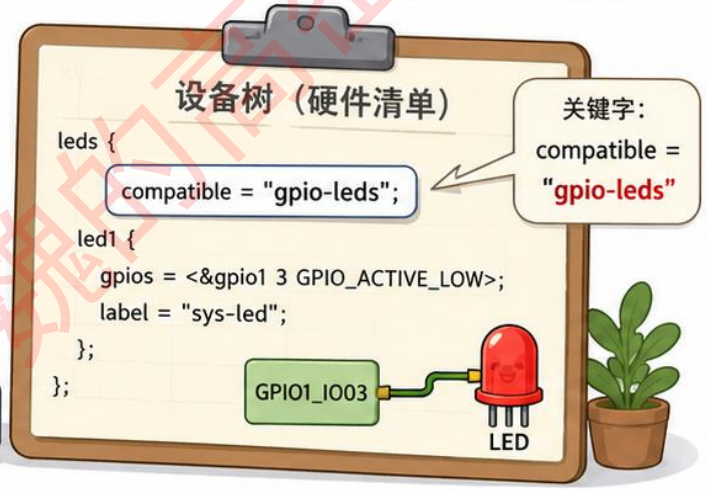


编译阶段

② Linux 内核：把驱动“编”进来



③ 设备树：告诉内核“板子上有什么硬件”



④ 运行阶段：匹配成功 → 驱动工作



系统启动后，Linux 内核查看设备树（硬件清单）

发现关键字："gpio-leds"

在内核中找到匹配的驱动：leds-gpio

驱动被调用，开始工作（配置 GPIO，控制电平）

实际 LED 硬件（点亮 / 闪烁）

一句话总结：

menuconfig 决定：内核里有没有这个驱动

设备树决定：这个驱动要不要在这块板子的这个硬件上工作

海贝博士的秘密小屋

dr-wei-arm@dr-wei-arm-linux: ~/linux/kernel_complete

Linux/arm 4.1.15 Kernel Configuration

us ---> (or empty submenus ----). Highlighted letters are hotkeys.
gend: [*] built-in [ ] excluded <M> module < > module capable

```
    General setup  --->
[*] Enable loadable module support  --->
[*] Enable the block layer  --->
    System Type  --->
    Bus support  --->
    Kernel Features  --->
    Boot options  --->
    CPU Power Management  --->
    Floating point emulation  --->
    Userspace binary formats  --->
    Power management options  --->
[*] Networking support  --->
    Device Drivers  --->
    Firmware Drivers  --->
    File systems  --->
    Kernel hacking  --->
    Security options  --->
-*- Cryptographic API  --->
    Library routines  --->
-*- Virtualization  ----
```

dr-wei-arm@dr-wei-arm-linux: ~/linux/kernel_complete

Device Drivers

-> (or empty submenus ----). Highlighted letters are hotkeys. Pre
[*] built-in [ ] excluded <M> module < > module capable

```
    ↑(-)
<*> MMC/SD/SDIO card support  --->
< > Sony MemoryStick card support  ----
    MXC support drivers  --->
[*] LED Support  --->
[ ] Accessibility support  ----
[ ] EDAC (Error Detection And Correction) reporting  ----
[*] Real Time Clock  --->
[*] DMA Engine support  --->
[ ] Auxiliary Display support  ----
< > Userspace I/O drivers  ----
```

```
--- LED Support
<*>   LED Class Support
< >     LED Flash Class Support
      *** LED drivers ***
< >   LCD Backlight driver for LM3530
< >   LED support for LM3642 Chip
< >   LED driver for PCA9532 dimmer
<*>   Dr_Wei_LED Support for GPIO connected LEDs
< >   LED Support for N.S. LP3944 (Fun Light) I2C chip
< >   LED Support for N.S. LP5521 LED driver chip
```

海归博士的倾心分享

Open

Kconfig
~/linux/kernel_complete/drivers/leds

Save

```
179           To use a pin as gpio pca9532_type in pca9532_platform data needs to
180           set to PCA9532_TYPE_GPIO.
181
182 config LEDS_GPIO
183         tristate "Dr Wei LED Support for GPIO connected LEDs"
184         depends on LEDS_CLASS
185         depends on GPIOLIB
186         help
187           This option enables support for the LEDs connected to GPIO
188           outputs. To be useful the particular board must have LEDs
189           and they must be connected to the GPIO lines.  The LEDs must be
190           defined as platform devices and/or OpenFirmware platform devices.
191           The code to use these bindings can be selected below.
192
```

海贝博士Dr.魏 版权所有

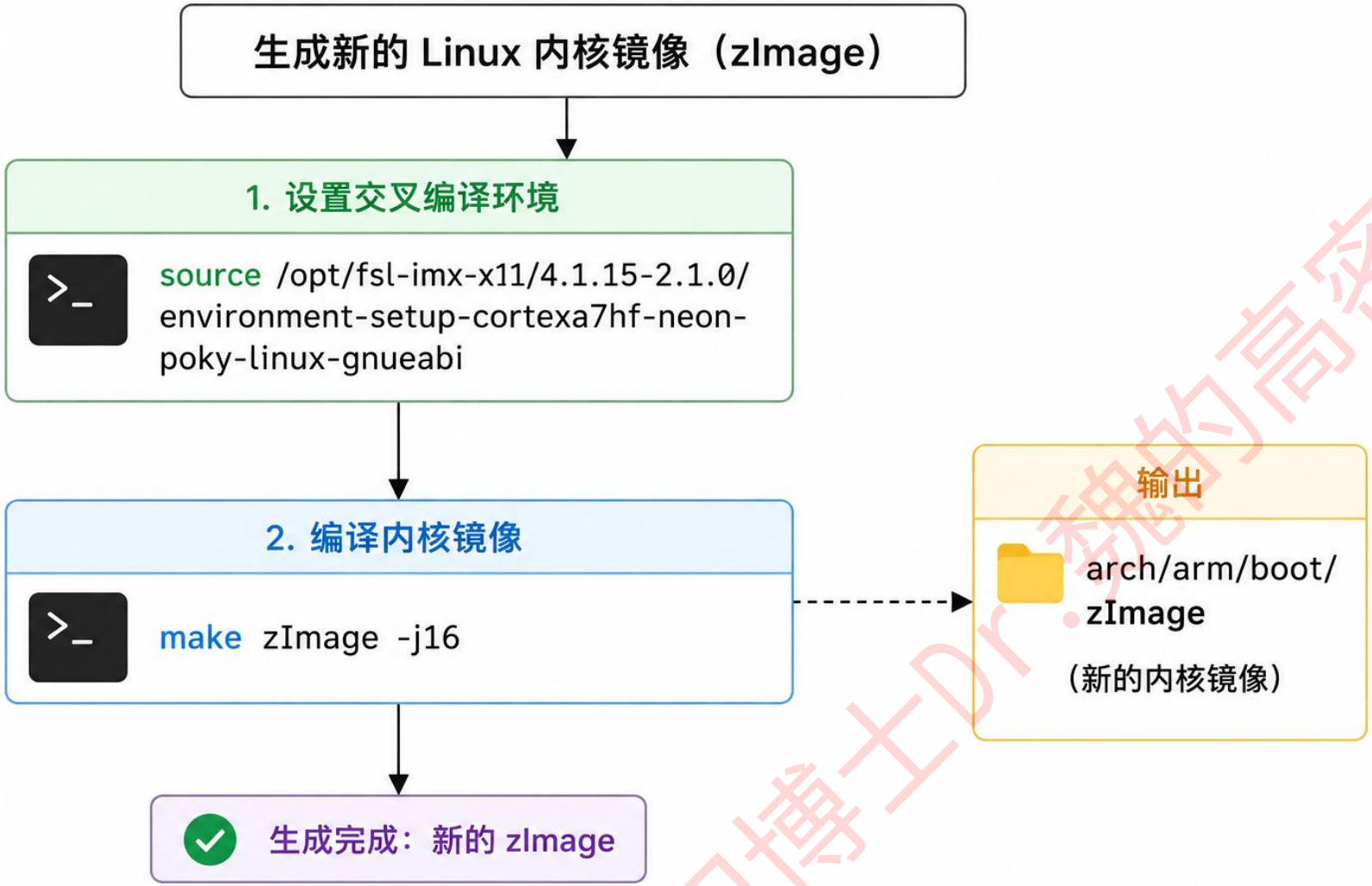
Open .config ~/linux/kernel_complete Save

```
3146 CONFIG_MXC_HDMI_CEC=y
3147 CONFIG_NEW_LEDS=y
3148 CONFIG_LEDS_CLASS=y
3149 # CONFIG_LEDS_CLASS_FLASH is not set
3150
3151 #
3152 # LED drivers
3153 #
3154 # CONFIG_LEDS_LM3530 is not set
3155 # CONFIG_LEDS_LM3642 is not set
3156 # CONFIG_LEDS_PCA9532 is not set
3157 CONFIG_LEDS_GPIO=y
3158 # CONFIG_LEDS_LP3944 is not set
3159 # CONFIG_LEDS_LP5521 is not set
3160 # CONFIG_LEDS_LP5523 is not set
```

Open Makefile ~/linux/kernel_complete/drivers/leds Save

```
19 obj-$(CONFIG_LEDS_COBALT_QUBE)          += leds-cobalt-qube.o
20 obj-$(CONFIG_LEDS_COBALT_RAQ)           += leds-cobalt-raq.o
21 obj-$(CONFIG_LEDS_SUNFIRE)              += leds-sunfire.o
22 obj-$(CONFIG_LEDS_PCA9532)              += leds-pca9532.o
23 obj-$(CONFIG_LEDS_GPIO_REGISTER)        += leds-gpio-register.o
24 obj-$(CONFIG_LEDS_GPIO)                 += leds-gpio.o
25 obj-$(CONFIG_LEDS_LP3944)               += leds-lp3944.o
26 obj-$(CONFIG_LEDS_LP55XX_COMMON)        += leds-lp55xx-common.o
27 obj-$(CONFIG_LEDS_LP5521)               += leds-lp5521.o
```

魏的博士IT·魏的高密乡



海归博士Dr.魏的高密乡

source /opt/fsl-imx-x11/4.1.15-2.1.0/environment-setup-cortexa7hf-neon-poky-linux-gnueabi

make zImage -j16

```
dr-wei-arm@dr-wei-arm-linux:~/linux/kernel_complete/drivers/leds$ ls
built-in.o          led-core.o          leds-cobalt-qube.c  leds-gpio.o
   leds-ss4200.c          leds-wrap.c
dell-led.c          leds-88pm860x.c     leds-cobalt-raq.c   leds-gpio-register.c
   leds-sunfire.c         led-triggers.c
Kconfig             leds-adp5520.c      leds-da903x.c       leds.h
c  leds-syscon.c          led-triggers.o
led-class.c         leds-asic3.c        leds-da9052.c       leds-hp6xx.c
   leds-tca6507.c         Makefile
led-class-flash.c   leds-bd2802.c       leds-dac124s085.c   leds-ipaq-micro.c
   leds-versatile.c       modules.builtin
led-class.o         leds-blinkm.c       leds-fsg.c          leds-lm3530.c
   leds-wm831x-status.c   modules.order
led-core.c          leds-clevo-mail.c   leds-gpio.c         leds-lm3533.c
   leds-wm8350.c          trigger
```

dr-wei-arm@dr-wei-arm-linux: ~/linux/kernel_complete

```
                                LED Support
---> (or empty submenus ----).  Highlighted letters are hotkeys.  Pr
d: [*] built-in  [ ] excluded  <M> module  < > module capable


    --- LED Support
    <*>   LED Class Support
    < >     LED Flash Class Support
          *** LED drivers ***
    < >   LCD Backlight driver for LM3530
    < >   LED support for LM3642 Chip
    < >   LED driver for PCA9532 dimmer
    <M>   Dr Wei LED Support for GPIO connected LEDs
    < >   LED Support for N.S. LP3944 (Fun Light) I2C chip
    < >   LED Support for N.S. LP5521 LED driver chip
```

Activities Text Editor Aug 31 12:31

Open .config
~/linux/kernel_complete

```
3151 #
3152 # LED drivers
3153 #
3154 # CONFIG_LEDS_LM3530 is not set
3155 # CONFIG_LEDS_LM3642 is not set
3156 # CONFIG_LEDS_PCA9532 is not set
3157 CONFIG_LEDS_GPIO=m
3158 # CONFIG_LEDS_LP3944 is not set
3159 # CONFIG_LEDS_LP5521 is not set
3160 # CONFIG_LEDS_LP5523 is not set
3161 # CONFIG_LEDS_LP5562 is not set
3162 # CONFIG_LEDS_LP8501 is not set
3163 # CONFIG_LEDS_LP8860 is not set
```

海归博士Dr·魏的高密乡

驱动已配置为 M，是否要执行这三条命令？

前提：驱动已配置为模块（CONFIG_XXX=m）

只需要模块

只想得到可加载的 .ko 模块文件

Step 1
```
source /opt/fsl-imx-x11/4.1.15-2.1.0/
environment-setup-cortexa7hf-neon-poky-linux-gnueabi
```

Step 2
```
make modules -j16
```
.ko 模块文件

输出：xxx.ko（可加载的模块文件）

还要更新内核镜像

内核源码有改动（如内建代码变更），或需要整体刷新内核

Step 1
```
source /opt/fsl-imx-x11/4.1.15-2.1.0/
environment-setup-cortexa7hf-neon-poky-linux-gnueabi
```

Step 2
```
make zImage -j16
```
生成内核镜像

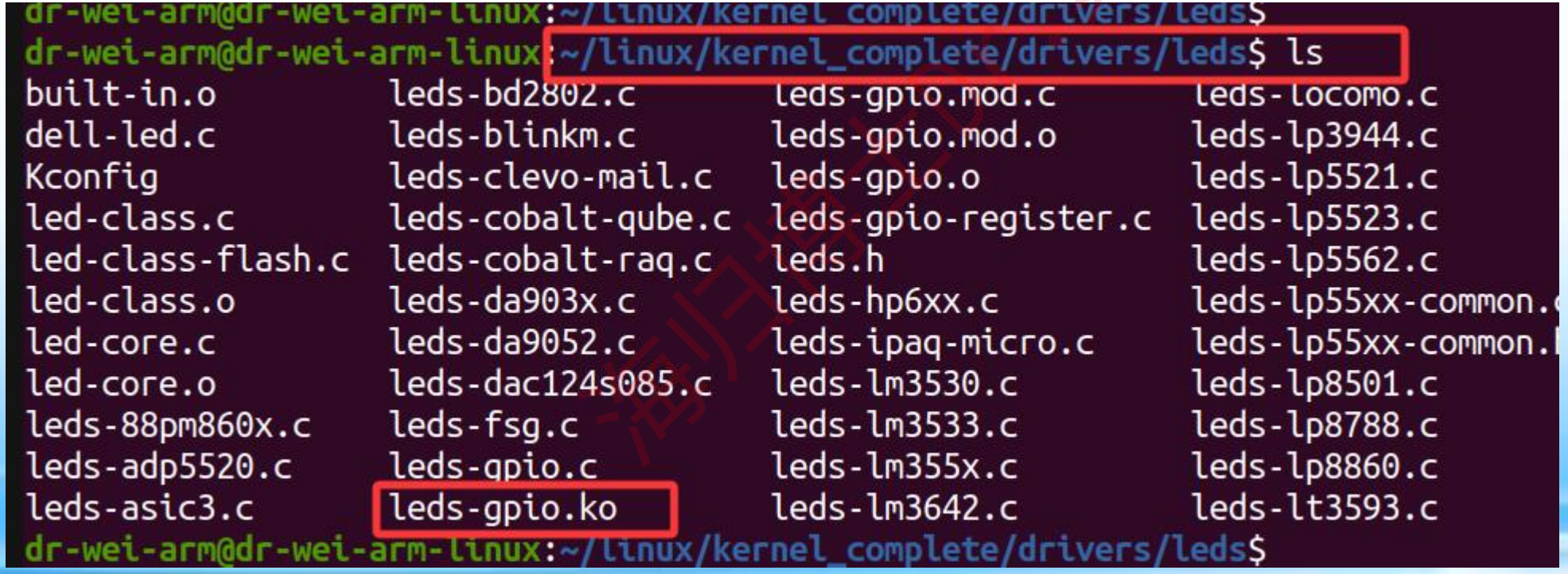
Step 3
```
make modules -j16
```
.ko 模块文件

输出：zImage（内核镜像）+ xxx.ko（模块文件）

结论：只编译模块：一般 2 步；同时更新内核镜像：再加 `make zImage -j16`

提示： 如果你只是把驱动改成 M，并想得到可加载的 .ko，重点是 `make modules`。

海归博士DT 魏的高含金


dr-wei-arm@dr-wei-arm-linux:~/linux/kernel_complete/drivers/leds$
dr-wei-arm@dr-wei-arm-linux:~/linux/kernel_complete/drivers/leds$ ls
built-in.o leds-bd2802.c leds-gpio.mod.c leds-locomo.c
dell-led.c leds-blinkm.c leds-gpio.mod.o leds-lp3944.c
Kconfig leds-clevo-mail.c leds-gpio.o leds-lp5521.c
led-class.c leds-cobalt-qube.c leds-gpio-register.c leds-lp5523.c
led-class-flash.c leds-cobalt-raq.c leds.h leds-lp5562.c
led-class.o leds-da903x.c leds-hp6xx.c leds-lp55xx-common.c
led-core.c leds-da9052.c leds-ipaq-micro.c leds-lp55xx-common.h
led-core.o leds-dac124s085.c leds-lm3530.c leds-lp8501.c
leds-88pm860x.c leds-fsg.c leds-lm3533.c leds-lp8788.c
leds-adp5520.c leds-gpio.c leds-lm355x.c leds-lp8860.c
leds-asic3.c leds-gpio.ko leds-lm3642.c leds-lt3593.c
dr-wei-arm@dr-wei-arm-linux:~/linux/kernel_complete/drivers/leds$

```
root@ATK-IMX6U:~#
root@ATK-IMX6U:~# cd /sys/class/leds
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds# ls
mmc0::  mmc1::
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds# cd
root@ATK-IMX6U:~# ls
beep  driver  led  leds-gpio.ko  shell
root@ATK-IMX6U:~#
root@ATK-IMX6U:~# insmod leds-gpio.ko
[  100.062691] fsl-asoc-card sound-es8388: priv->codec_priv.mclk_freq
[  100.082837] fsl-asoc-card sound-es8388: DPCM DAI Links only if ASR
root@ATK-IMX6U:~# [  100.107629] fsl-asoc-card sound-es8388: ASoC: CO
[  100.115744] fsl-asoc-card sound-es8388: snd_soc_register_card fail

root@ATK-IMX6U:~# cd  /sys/class/leds/
root@ATK-IMX6U:/sys/class/leds#
root@ATK-IMX6U:/sys/class/leds# ls
beep_dr_wei  mmc0::  mmc1::  sys-led_dr_wei_new
root@ATK-IMX6U:/sys/class/leds#
```

Linux 内核编译：驱动从 .c 到 zImage / .ko 的过程示意图

1. 配置为内置（编译进内核）

CONFIG_XXX=y

驱动源码
xxx.c
例如：
drivers/
leds/
leds-gpio.c

→ 编译 →

目标文件
xxx.o
例如：
drivers/leds/
leds-gpio.o

→ 加入内核对象文件集合 →

大量的 .o 文件（内核各部分）
arch/arm/xxx.o
mm/xxx.o
fs/xxx.o
drivers/xxx.o
…
drivers/leds/
leds-gpio.o

特点：
- 驱动在内核启动时就已存在
- 不需要 insmod / modprobe
- 成为 zImage 的一部分

包含内容：
- 内核核心代码
- 所有内置的驱动（=y）
- 初始化代码等

链接（Link）
将所有 .o 链接成 vmlinux
（未压缩的内核镜像）

↓

压缩 / 封装
生成可启动的内核镜像

↓

zImage
arch/arm/boot/zImage
（可在 Bootloader 中加载启动）

2. 配置为模块（编译成模块）

CONFIG_XXX=m

驱动源码
xxx.c
例如：
drivers/
leds/
leds-gpio.c

→ 编译 →

目标文件
xxx.o
例如：
drivers/leds/
leds-gpio.o

→ 生成模块 →

内核模块
xxx.ko
例如：
drivers/leds/
leds-gpio.ko

特点：
- 驱动不会出现在 zImage 中
- 需要在系统启动后通过 insmod / modprobe 加载
- 以 .ko 文件形式存在于根文件系统中

系统启动后
通过 insmod / modprobe 加载模块

↓

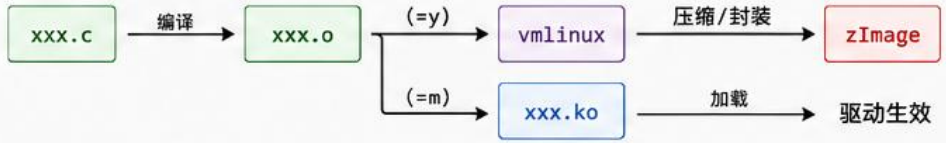
模块生效，驱动工作

.ko 文件存放位置
通常在：
/lib/modules/
$(uname -r)/
/kernel/...

总结对比

配置选项	编译结果	最终去向	加载时机
CONFIG_XXX=y	.c → .o → 链接进 vmlinux	zImage	内核启动时
CONFIG_XXX=m	.c → .o → .ko	xxx.ko	系统启动后手动/自动加载

整体流程图（简化）



海归博士Dr·魏的高密乡

make ARCH=arm menuconfig

make ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf- dtbs

imx6ull-14x14-emmc-7-1024x600-c.dtb

海贝博士Dr.魏的高密乡

source /opt/fsl-imx-x11/4.1.15-2.1.0/environment-setup-cortexa7hf-neon-poky-linux-gnueabi

make zImage -j16

make modules -j16

海归博士Dr·魏的高密乡

海归博士Dr·魏的高密乡

海归博士Dr·魏的高密乡

海归博士Dr·魏的高密乡

海归博士Dr·魏的高密乡

海归博士Dr·魏的高密乡

海归博士Dr·魏的高密乡